Context Dependent Pattern Recognition - A Framework for Hybrid Architectures Bridging Chaotic Neural Networks Based on Recursive Processing Elements and Symbolic Information

Emilio Del-Moral-Hernandez, Humberto Sandmann, and Gleison Araújo

Abstract—**This work discusses a hybrid structure that conjugates connectionist associative memories and deterministic automata, for the implementation of context dependent pattern recognition. The associative component of the hybrid system is built through coupled recursive maps with bifurcation and chaotic dynamics (Recursive Processing Elements - RPEs). Its output feeds a deterministic state machine that controls the context of the pattern recognition tasks and produces related symbolic outputs. The proposal is illustrated in a scenario for context dependent (visual) pattern recognition, performed by an autonomous agent. Such "Learner" agent alternates between contexts of unsupervised image recognition and contexts of interaction with a "Teacher" agent, in supervised sections of image recognition. Computational experiments and related measures show the effectiveness of the proposal.**

I. Introduction

THE increasing presence of different aspects of dynamics in novel models of neurocomputing is one of the trends that we can observe in the communities concerned with the modeling of neural function and the development of more complex tools for neurocomputing. As some of the examples of this trend, we can mention spiking model neurons, attractor networks, chaotic neural networks, and the use of feature extraction techniques employing time-frequency analysis and other techniques in which the emphasis on events in time is present [1]-[8]. In this scenario, model neurons with rich dynamics, including the representation of the phenomena of bifurcation and cascading to chaotic behavior, as observed in real neurons, are an important area in neurocomputing and neural modeling research [3]-[10].

In the context of the modeling of cognition, recognition, behavior, action, and collective behavior, among others, in addition to the microscopic aspects of dynamics naturally present in the above mentioned neural models, it is important to additionally consider other relevant modeling elements involving more macroscopic time scales such as the ones observed in dialog processes, in general interactions between biologic autonomous entities, as well as the dynamics of interaction between an autonomous entity and its changing surroundings.

This paper touches aspects of dynamics at the two levels mentioned above. From one side, more related to the microscopic level and the neurodynamics, the paper addresses chaotic neural networks based on coupled recursions that organize in ordered oscillatory states representing recognized information and can autonomously change to disordered chaotic behaviors during the processes of search for previously learned information and recognition of stimulus patterns [6], [7], [9], [10]. The networks of coupled recursive nodes here studied, named RPEs architectures, with RPEs being a short form for Recursive Processing Elements, are able to represent relevant stimulus through collective spatiotemporal periodic attractors, as well as to perform noise immune recognition of previously learned patterns, based on the exploration of the basins of attraction of the periodic states that represent the relevant stored information [6], [7], [9], [10].

One of the problems that we frequently face in the application of many connectionist models is the lack of embedded bridges to symbolic models, which, if available, could make easier the task of exploring the large amount of techniques that we have developed in the arena of digital computing, such as high level programming, finite state machine design, deterministic and stochastic automata, and many different modeling techniques based on digital computing, which are well developed and documented [11], [12]. Trying to partially address this need, this paper has as one of its contributions the development of a hybrid structure which joins two components: 1) The operation of connectionist associators based on the already mentioned coupled recursive maps and representation of information through periodic attractors, which will be responsible for the recognition of relevant stimulus with high robustness to noise [2], [6], [7], [9], [10], and 2) The operation of a discrete state and discrete time automata [12] which is driven by the information produced by the RPEs associator, and , at the same time, impacts on its behavior by controlling the contexts under which such associator will recognize patterns.

With that hybrid approach, we have the goal of allowing a bridge of the RPEs associators to symbolic information and the related temporal aspects in macroscopic time scales, such as, for example, time frames for dialog-like interactions between autonomous agents that exchange symbolic information through their perceptual channels. We explore

Emilio Del-Moral-Hernandez is with Department of Electronic Systems Engineering, Polytechnic School of the University of Sao Paulo, Brazil, (e-mail: emilio_del_moral@ieee.org).

Humberto Sandmann is with Department of Electronic Systems Engineering, Polytechnic School of the University of Sao Paulo, Brazil. His Ph.D. degree work was supported by CNPq, under grant number 140632/2007-7 (e-mail: hsandmann@ieee.org).

Gleison Araújo is with Department of Electronic Systems Engineering, Polytechnic School of the University of Sao Paulo, Brazil, (e-mail: gleisonsjc@yahoo.com.br).

thus a hybrid structure, blending connectionist and symbolic modules, employing chaotic neural networks and discrete time automata. Fig. 1 represents such kind of arrangement, joining the operation of associators based on coupled RPEs architectures and deterministic state machines.

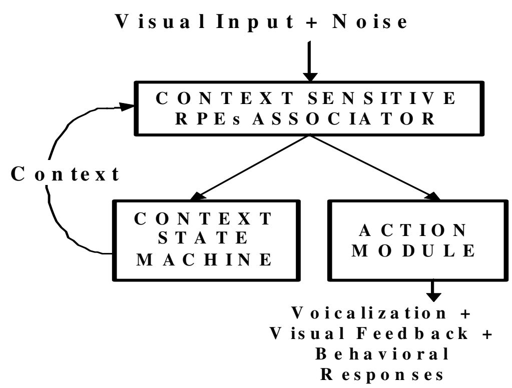

Fig. 1. Context sensitive RPEs associators interacting with a context state machine. The RPEs associator recognize patterns within a context dependent universe. By its turn, the patterns recognized by the associator affect the evolution of the context state machine.

The potential applications of this type of hybrid architecture include the modeling of dynamic context sensitive pattern recognition and association, as well as the ability to address the processing and recognition of time sequences, and the possibility of implementing and modeling closed loop systems where sensing and context switching can affect each other.

In the following section (II), we discuss the blend of order and chaos in pattern recognition and association in RPEs networks. In Section III, we discuss dynamic context sensitive association, and how this can be implemented through the interplay between RPEs associators and a discrete state machine. In Section IV, relevant aspects of asynchronous events and asynchronous input changes are addressed. In section V, we discuss simulation results on the modeling of context dependent pattern recognition, and Section VI has the Conclusion.

II. Blend of Order and Chaos in Pattern Recognition and Association in Networks Based on Recursive Processing Elements

For the associative segment of the hybrid structure here addressed, this paper explores architectures based on coupled recursive nodes that interact through parametric coupling and exhibit bifurcation and cascading to chaotic dynamics. Since Aihara's proposal of his model neuron with dynamics and chaotic behavior, many different efforts for the modeling of the rich dynamics of real neurons and many different models for the implementation of associative memories based on nodes with rich dynamics have been developed, sometimes reaching more complex functionalities with respect to models based on the more traditional dynamicless model neurons, or, alternatively, reaching higher performance in different aspects [3]-[7], [9], [10]. In addition, these models also help in strengthening the ability of artificial neural networks to mimic complex phenomena observed in real neurons, since they provide elements for the representation of some aspects of complex neurodynamics.

The RPEs models were developed under this scenario, of exploring rich dynamics for enhancing neural computation and neural modeling. The RPEs architectures here explored are connectionist associative structures and have been used in other works for the implementation of associative memories, heteroassociative memories, and also as a framework for the coupling of spiking model neurons with bifurcation and chaotic dynamics [2], [6], [7], [9], [10].

Fig. 2 represents generically an RPEs associative structure. In the remaining of this section, we summarize the basic concepts involved in the operation of RPEs associative networks, including the definition of connection weights and the descrition of the operation of RPEs networks during the process of pattern recovery and pattern association.

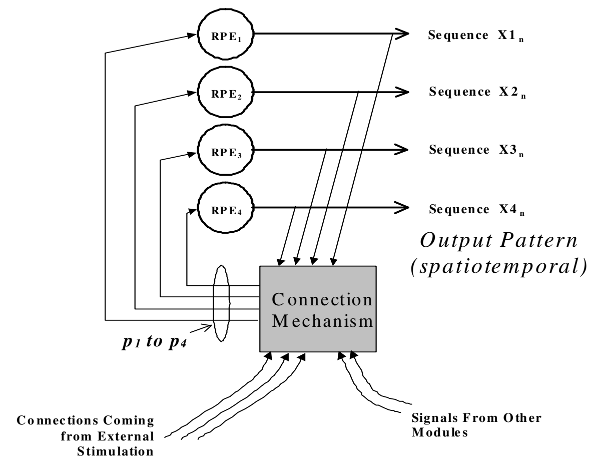

Fig. 2. A fully connected associative network composed of 4 RPE nodes. Each circle RPE_1 to RPE_4 represents one bifurcating logistic recursion. The connection among nodes is promoted through proper changes in bifurcation parameters of the RPE nodes (p_1 to p_4). These changes are driven by the internal assembly state, input signal paths and inter assembly connection paths. Figure adapted from [6].

In the modeling and computational experiments discussed in later sections, we consider an illustrative problem dealing with context dependent image recognition tasks. For image recognition and image representation problems, each pixel of the visual field is represented by one RPE node with oscillatory behavior. The binary pixel value (black or white) is coded through the phase of oscillation of a period-2 oscillatory trajectory performed by the $\boldsymbol{x}$ state variable of the RPE node., between two values α and β. Such period-2 cycles performed by the RPE have two distinct phases of oscillation (first α then β, or first β then α), what is sufficient to represent binary values, such as +1 and -1, or black and white [6]. Equation (1) shows the evolution of the logistic recursive map [13], which is used in these experiments to define the state variable $\boldsymbol{x}$ of a node $\boldsymbol{i}$ composing an RPEs associator. In (1), $\boldsymbol{n}$ is discrete time, $x_{i,n}$ and $x_{i,n+1}$ are consecutive values of the state variable for node $\boldsymbol{i}$, and the $\boldsymbol{p_{i,n}}$ is the bifurcation parameter of the logistic recursion.

$$x_{i,n+1} = p_{i,n} \,.\, x_{i,n} \,.\, (1 - x_{i,n}) \tag{1}$$

Fig. 3 shows the bifurcation diagram for the logistic map, which presents (in the long term) fixed-point behavior for $\boldsymbol{p}$ values bellow $\boldsymbol{3}$, period-2 trajectories for $\boldsymbol{p}$ values in the range $\boldsymbol{3}$-to-$\boldsymbol{3.45}$, cyclic trajectories with larger periods for $\boldsymbol{p}$ values in the range $\boldsymbol{3.45\text{-}3.56}$, and chaotic trajectories for $\boldsymbol{p}$ values above $\boldsymbol{3.56}$ [4], [13]. The excursion of the $\boldsymbol{p_i}$ parameters in networks of RPEs are limited from above to $\boldsymbol{p_{max} = 3.8}$ (this value of $\boldsymbol{p}$ produces chaotic trajectories in the state variables $\boldsymbol{x}$), as well as from below, to $\boldsymbol{p_{min} = 3.25}$ (this value of $\boldsymbol{p}$ produces the period-2 oscillations in the state variables $\boldsymbol{x}$, for the coding of the stored binary pixels).

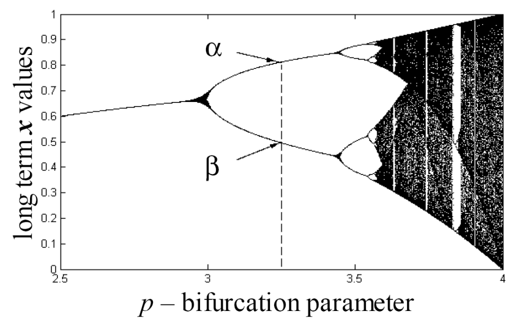

Fig. 3. Bifurcation diagram for the logistic recursion: $\boldsymbol{x_{n+1} = p \,.\, x_n \,.\, (1\text{-}x_n)}$, including the indication of one specific periodic long-term trajectory, formed by the oscillation between two values $\boldsymbol{\alpha \cong 0.81}$ and $\boldsymbol{\beta \cong 0.5}$. This period-2 cycle, which emerges for $\boldsymbol{p = 3.25}$, can be used for the representation of a single pixel in a binary image.

The visual field considered here in the image recognition experiments is assumed to have 10x10 pixels, what defines the size of the RPEs associator: 100 nodes. Equation (2) defines the synaptic weight value $\boldsymbol{w_{ij}}$, linking node $\boldsymbol{j}$ to the node $\boldsymbol{i}$.

$$w_{ij} = \sum_{\mu=1:M} (I_{\mu,j} \,.\, I_{\mu,i}) \tag{2}$$

In (2), the 100 pixels vector $\boldsymbol{I_\mu}$ represents one of the images stored in the RPEs associator, $\boldsymbol{\mu}$ identifies that image in the full set of $\boldsymbol{M}$ relevant images, and $\boldsymbol{i}$ and $\boldsymbol{j}$ identify two different pixels of the visual field (pixels $\boldsymbol{i}$ and $\boldsymbol{j}$ in the image), as well as the two nodes that represent those pixels. As in Hopfield associative memories [14], the connection weights $\boldsymbol{w_{ij}}$ defined as above are calculated through the sum of outer products of the $\boldsymbol{M}$ images, using bipolar representation (-1 and +1, instead of 0 and 1) for the binary pixels. Equations (3a) and (3b) represent the parametric coupling between RPE nodes. They define how those recursive nodes affect each other through changes of their bifurcation parameters $\boldsymbol{p_i}$, what can sometimes drive individual nodes to ordered period-2 trajectories that represent the binary pixels, while at other moments can drive individual nodes to chaotic trajectories, aiming to facilitate the search for stored images which are consistent with the received visual stimulus..

$$d_i = -\,(x_i - k_1).(\Sigma_j\ \ w_{ij} \,.\, (x_j - k_1)) + k_2 \tag{3a}$$

$$\Delta p_i = d_i \,.\, c - e \tag{3b}$$

In (3), the discrete time $\boldsymbol{n}$ was omitted for simplicity of notation, $\boldsymbol{d_i}$, $\boldsymbol{x_i}$, $\boldsymbol{x_j}$, and $\boldsymbol{\Delta p_i}$, all change with time, while $\boldsymbol{w_{ij}}$ **,** $\boldsymbol{k_1}$, $\boldsymbol{k_2}$, $\boldsymbol{c}$, and $\boldsymbol{e}$, are constant values. The two essential roles of equations (3) in the operation of the RPEs network are: 1) to promote complex dynamics (higher values of $\boldsymbol{p_i}$) when activities $\boldsymbol{x_i}$ and $\boldsymbol{x_j}$ are inconsistent with the expected correlation $\boldsymbol{w_{ij}}$; 2) to promote more ordered dynamics (lower values of $\boldsymbol{p_i}$) when activities $\boldsymbol{x_i}$ and $\boldsymbol{x_j}$ are in good agreement with the expected correlation $\boldsymbol{w_{ij}}$ [6].

In order to have those two desired behaviors, the changes in the bifurcation parameter of a given node $\boldsymbol{i}$ are conducted by a global disagreement indicator $\boldsymbol{d_i}$, as expressed in (3a). This global disagreement for the node $\boldsymbol{i}$ takes into consideration all the activities $\boldsymbol{x_j}$ of the remaining nodes $\boldsymbol{j}$, and it can be used to drive the local bifurcation parameter $\boldsymbol{p_i}$, as it is expressed in (3b). In this way, the term $\boldsymbol{\Delta p_i}$ in (3b) promotes gradual changes in the local bifurcation parameter $\boldsymbol{p_i}$ of node $\boldsymbol{i}$. In this scenario of planed changes of the $\boldsymbol{p_i}$ parameters, the constant $\boldsymbol{c}$ implements the matching of the scale of the disagreement $\boldsymbol{d_i}$ with the scale of the gradual changes $\boldsymbol{\Delta p_i}$, $\boldsymbol{e}$ works as a dumping constant that helps $\boldsymbol{p_i}$ to move to lower values (ordered dynamics), the constant $\boldsymbol{k_1}$ is the average value of the $\boldsymbol{x}$ variables, and $\boldsymbol{k_2}$ is an offset for the disagreement measure $\boldsymbol{d_i}$.

To close this section on the concepts of RPEs networks, we summarize in Table I one of the interesting results obtained in previous works with the RPEs auto-associative architectures [9]: Table I shows the larger immunity to noisy promptings with respect to traditional Hopfield associative networks.

TABLE I
PERFORMANCE CONTRAST FOR HOPFIELD ASSOCIATIVE MEMORIES AND RPES ARCHITECTURES

Noise Level	Average Hamming Error in Hopfield Networks	Average Hamming Error in RPEs Networks
10%	0.00%	0.00%
20%	0.33%	0.18%
30%	2.12%	1.26%

Each row shows the average error in the recovered images, for an amount of input noise as specified in the first column [9].

III. CONTEXT SENSITIVE ASSOCIATION, AND THE INTERPLAY BETWEEN RPES ASSOCIATORS AND DISCRETE STATE MACHINES, FOR CONTEXT SWITCHING MODELING

Associators and heteroassociators such us the ones described in the previous section [2], [6], [7], [9], [10] and similar associators based on Hopfield and Kosko models [14], [15] can be appropriate for several computing tasks such as pattern recovery through auto-associative memories and one to one bidirectional associations, under simple scenarios having a single fixed set of relevant patterns. In several other scenarios though, a more flexible form of association, having dynamic rules of operation and dynamic set of target patterns, which can change according to the previous history of stimulus or according to the evolution in time through different contexts of perception, action, dialog and behavior, can be necessary. That is the case for example for the processing and recognition of sequences, as well as for the modeling of association or recognition tasks which are sensitive to changes in the focus of attention. It is also the case of somehow similar scenarios, where context and focus of attention are driven externally with respect to the perception system, by signals coming from higher level information processing modules [11], [16].

With the above considerations in mind, the goal of this section is to develop the main concepts of a framework for the modeling of context sensitive pattern recognition, through the modulation of the connections matrix of the RPEs associator, for the production of dynamic multi-context associations.

The possibility of having extra synaptic inputs in RPEs architectures, in addition to the stimulation inputs and the typical feedback inputs of auto associative architectures, such as the extra inputs represented in Fig. 2, with the label "Inputs from other modules" [6], can be of some use for this purpose, although subject to produce limited results. Such approach of having extra synaptic inputs to the nodes of the associative structure, in order for these extra inputs to carry the context information, suffers the limitation of the linear composition between the external control path and the feedback path, as a result of the linear combination of inputs in typical synaptic dendritic trees. In practice, this linear nature of the composition of nodes inputs reduces the impact that the context signals can have on the structure of the performed associations. In other words, such a control mechanism cannot change largely the landscape of attractors of the RPEs associators, what is sometimes necessary when we consider marked changes of context. More effective results can be produced by implementing mechanisms in which the signals carrying the definition of the current context have direct impact on the associator's synaptic connections matrix, allowing thus a more direct change of the landscape of attractors. This approach, i.e., the direct modulation of the synaptic connections of RPEs associators, was the one adopted in this study.

Once defined the way the context sensitive association is implemented in the RPEs structure (concretely, through synaptic modulation), we can now move to the description of how the dynamic context signals are generated. An effective way to implement a driver signal for the control of the multi-context associator, which at the same time informs the present context and summarizes the previous history of the stimulation and the evolution of the associator through time, corresponds to using an state machine [12]. That state machine can receive the output of the associator and can retain information regarding the present state, which is a result of the previous evolution of the whole structure (RPEs associator plus the deterministic automata). It also can, in response to specific inputs, produce the planned next state of the automata and, related to such next state, the signals for the driving of the future context sensitive associations.

In the experiments used to illustrate the hybrid technique described in this paper, we considered a simple scenario with context sensitive recognition, involving only three relevant contexts, named here A, B, and C. Each one of these contexts is related to different image recognition tasks considered in the modeling of an autonomous agent, which is able to receive visual stimulation in the three different contexts, perform the identification of such visual input, and act specifically in response to it and accordingly to the current context. Such autonomous agent will be named here "Learner", and it interacts with the environment in certain circumstances (i.e., when in Context A), receiving visual inputs from it. In other circumstances (when in Contexts B and C), it also interacts with a second autonomous agent, named here "Teacher". This second agent is not the main object of the modeling being done here, which targets context sensitive association. Nevertheless, similar techniques as the ones used for the modeling of the Learner agent could in principle be employed for both of them.

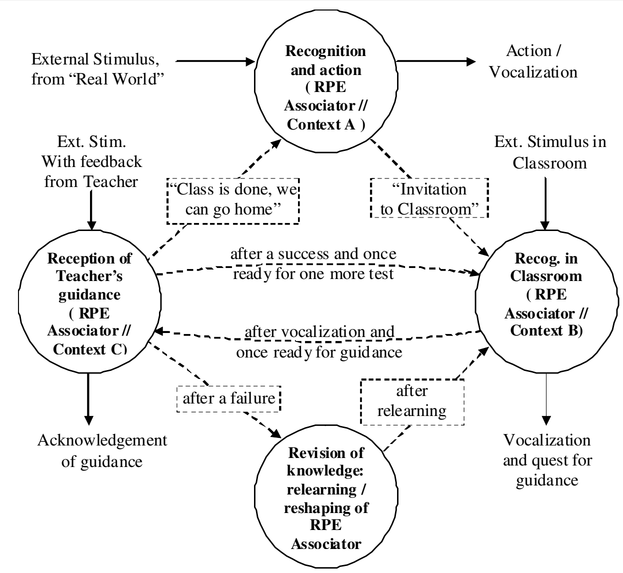

Fig. 4. Representation of the 3 contexts of association represented in the context state machine, Context A, Context B and Context C, as well as representation of the recognition events that promote the changes among different contexts. .

Metaphorically, we can say that Context A regards image recognition during "real life" of the Learner agent, i.e., when it is at "home", or, in general, when it is "outside learning sections at school". Similarly, we can also say that Contexts B and C regard the image recognition actions performed by the Learner during the training sections at school. These concepts are developed in the next paragraphs, which explain in more detail the nature of Contexts A, B, and C.

Context A, as mentioned above, regards the recognition tasks performed regularly by the Learner. In our simulations, such Context A targets images of shapes, such as "A Square", "A Round Shape", "A Star", and so on. The images used in our experiments for the representation of these shapes and the representation of other relevant visual elements to be recognized by the Learner can be found in

Fig. 6 and Fig. 7, in section V, together with the discussion of computational experiments. Under context A, in addition to the shape forms, the Learner should also be able to recognize a specific visual input that tells that a "training section at school" is needed. This visual stimulus is implemented here through an image including a chair and a classroom desk (last image in Fig. 6), and it can be produced by Teacher, by another neighbor agent, by an activities day schedule, or even by any self warning mechanism available to the Learner, indicating the convenience of a training section at school. When the recognition of such visual input happens, the autonomous agent responds to this "Let us learn!!" invitation, by producing some kind of feedback (vocalization, for example), and by switching its current context from Context A to Context B (and eventually to Context C, as explained ahead). This switching of context also corresponds to the change to the classroom environment, where Contexts B and C are the relevant ones.

In Context B, the recognition of the same shape forms as the ones considered in Context A is performed by the Learner, with the difference that now they happen "in classroom", under the supervision of the Teacher. Several shape stimuli are presented to the Learner, which tries to do the identification correctly and then vocalizes its result. After each individual recognition is performed by the Learner, it switches to Context C, in order to receive Teacher's feedback.

Context C supports the communication between Teacher and Learner during a training section in classroom. In context C, the targets of the visual recognition task are not the shape images, which were important for Contexts A and B. Instead of those, the targets are visual signals produced by the Teacher, and they carry some advice to the Learner. More concretely, the targets of the recognitions performed under Context C are Teacher's visual messages that tell the Learner about a success (a visual signal such as the image of a smile, like the first image in Fig. 7) or a failure (a slanted cross indicating a mistake, like the second image in Fig. 7). These guidance images inform the Learner what was the result of the supervised recognition of a shape form, which was previously tried by the Learner, just a bit before entering Context C, when still under Context B.

After the visual feedback from the Teacher is processed by the Learner (Context C), the current context is switched back to Context B, for a new shape recognition task, which will be done as before, under the supervision of the Teacher. This process of alternation between Contexts B (recognition of a shape form) and C (recognition of the visual guidance from Teacher) repeats for several sections. Eventually, the "Teacher feels happy" with the results of the class, and produces a different visual guidance, which, instead of conveying failure or success messages, tells the Learner that the training section is over, and that the Learner "Can go home". This is being done with the image of a house (last image in Fig. 7).

Fig. 4 shows the representation through circles of the contexts of association A, B and C just described, as well as it shows the dynamics of transitions among them (dashed arrows), the recognition events that promote the changes among the three contexts (on the dashed arrows) and the flow of input information received by the RPEs associators in each context and the produced action (solid arrows).

IV. Asynchronous Evolution of Context Switching and Asynchronous Input Changes

An important issue for the proper operation of the hybrid structure based on the RPEs connectionist associator and the Context state machine described in Fig. 4 regards the synchronization of both subsystems. We will thus make some considerations on aspects related to the time scale and nature of the time evolution of the two parts of the hybrid system, before we move to the discussion of experimental results.

From the point of view of the RPEs architecture described in Section II, we have the need for a fine time marker which paces the evolution of the logistic recursive map in each RPE node. This time marker, whose time scale we could call microscopic, relates to the local evolution of the recursive nodes, according to the equations discussed in Section II - (1), (3a), and (3b).

This microscopic time scale, although appropriate for the dynamical processes in the RPEs associator, does not reflect properly the relevant dynamics around complete actions of pattern recognition: a large number of the microscopic time steps used by the recursive maps have to take place before the completion of a single cycle of information retrieval and before we have the formation of a stable cyclic spatiotemporal patterns emerging from the structure of coupled RPEs. We have thus the need for a second relevant time scale, which we could call "macroscopic time scale", which is more related to complete cycles of pattern recovery, as well as to the transitions of the state machine represented in Fig. 4. In addition to this difference of magnitude between the time scales of the two components of the hybrid system, we observe that the volume of microscopic time steps that are involved in the recovery and stabilization cycle that define one step in the macroscopic time scale is not fixed. This volume of microscopic time steps changes according to the level of noise in the input pattern, to the previous state of the RPEs network, to the number and the repertoire of stored patterns, and according to many other specificities of the RPEs architecture, its training set and its previous history. As a consequence, the macroscopic time markers do not happen at regular intervals of the microscopic time scale. This requires the existence of a mechanism that autonomously generates triggers for the context state machine, whenever a new pattern is asynchronously produced by the context sensitive RPEs associator. We can produce that trigger based on the settling of the RPEs structure, given that the recovery of a spatiotemporal pattern is indicated by the generalized decrease of the values in the bifurcation parameters $\boldsymbol{p_i}$ of the RPE nodes, together with the trend of the nodes dynamics to perform more ordered trajectories.

Another issue related to asynchronous operation of the several elements involved in the hybrid structure here studied and the relevant stimuli events produced by the environment and by other autonomous agents, which are in fact the sources of image stimulus for the autonomous agent modeled by the hybrid structure implementing context sensitive association, refers to the production of asynchronous promptings to the associator stage. This aspect of asynchronous interaction between associator and external elements can be addressed with the inclusion of a novelty detection stage, at the input pattern path. This was the approach used here, where a simple threshold based input novelty detector was employed for the triggering of the associator module. In this way, the triggering of pattern recognition and association and the triggering of the "moves" happening in the context state machine are autonomously driven by the stimulus and its significant changes. Notice that this provides some basic elements for future explorations targeting the dealing with time warping in spatiotemporal pattern recognition.

To conclude this section on aspects of asynchronous operation of several relevant elements of our study, we want to discuss some related aspects in what regards the computational implementation of the previously described models, taking in consideration such aspects of diverse time scales and asynchronous events.

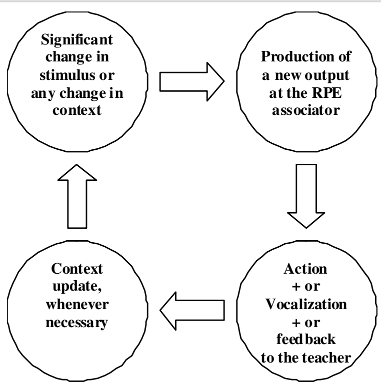

Fig. 5. Four consecutive phases considered in a complete cycle of associative task and related resultant actions.

The matching between the particular time evolutions of the several individual elements involved in the hybrid structure and the surrounding environment can be facilitated by considering an intermediary time scale between the macro and micro time scales, which we could call mesoscopic time scale. This intermediate time scale regards the different phases of each cycle of interaction between the autonomous agent and the external elements. As illustrated in Fig. 5, four consecutive phases happening in this mesoscopic time scale are considered as composing a complete cycle of associative task and related resultant actions: 1) Detection of significant change in input stimulus or in the current context; 2) Association task; 3) Action in response + Vocalization or similar feedback, if relevant; 4) Context update, if needed. Such break of each interaction agent-environment cycle into several components was found to be useful not only for computational implementation purposes, but also for conceptual and modeling purposes.

V. Simulation Results on Modeling of Context Dependent Pattern Recognition

This section presents the details and the results of computational experiments carried out for the implementation and study of the modeling scenario described conceptually in Sections II to IV, involving the two autonomous agents, the Learner and the Teacher, image recognition tasks sensitive to context, performed by RPEs associators, and the consideration of three relevant contexts: A (shape recognitions outside class), B (supervised shape recognitions in class), and C (recognition of the Teacher signs commenting on the results of the Learner in class). For the evaluation of the robustness of the context sensitive associations and the interaction Learner-Teacher, the computational experiments consider different levels of random noise (from 0% to 50%) added to the images presented to the Learner.

Fig. 6 shows the set of input images which are relevant for the contexts A and B. These images mainly correspond to different geometric shapes (a Triangle, a Round Form, a Squared form, a Cross, and a Star). The last image of this set in Fig. 6, is not a geometric shape, and it has the special role of promoting the switching from Context A to Context B: as described in Section III, such specific image in this set corresponds to an invitation to the classroom environment. Such image is labeled as 'Let us learn' in Fig. 6.

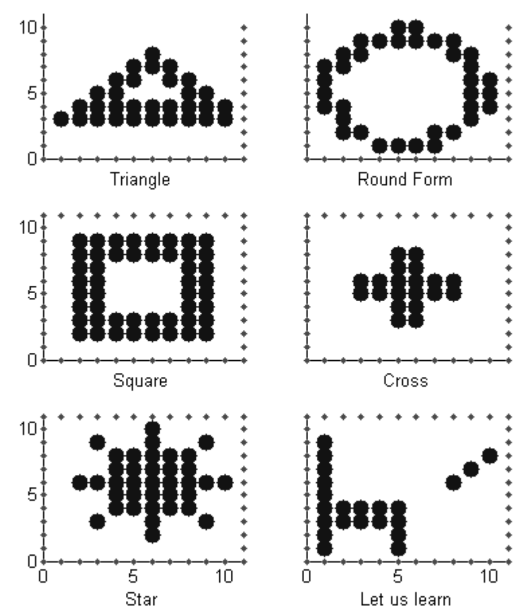

Fig. 6. Set of input images relevant for contexts A and B (recognition of images in real world, and recognition in classroom, respectively). Five of these images correspond to different geometric shapes, and the sixth one corresponds to a sign to "Let us learn", which is used to invite to a training section in "classroom".

For the simulations carried out, we considered the shape images of Fig. 6 presented in random order, with an equal probability of 16% for each one of the 5 shape symbols. The remaining 20% probability was reserved for the image corresponding to the invitation “Let us learn”. This relatively high probability of this sixth image ensures frequent transitions to the Context B, allowing the study of the processes of image recognition in the classroom environment (Contexts B and C).

Fig. 7 shows the set of input images relevant for Context C, i.e., recognition of the guidance images used by the Teacher, when commenting on the Learner’s performance. The first image in Fig. 7 conveys approval from the Teacher through a smile, the second image indicates, through a slanted cross, a mistake by the Learner done in the last shape recognition task performed in class, and the third image indicates that the Teacher is saying “Learning is done for today, you can go home”.

The labels under each guidance image in Fig. 7 illustrate their meanings and suggest possible phrases of an implicit dialog being carried out through the images: “Good Job ... let us move to another test with shapes ...”; “Wrong Answer ... let us re-study things”; “Learning is done for today, you can go home now”. In fact, such phrases under the images are good candidates for vocalization associated to the interaction of the agents Teacher and Learner.

For this second set of images, we considered that 90% of the images presented to the Learner agent during Context C correspond to Teacher comments, either indicating a success or indicating a failure (the specific image being a smile or a slanted cross will depend on the success or failure of the Learner, and, of course, the splitting between these two cases is not a process to be described by probabilities, but is instead a deterministic action taken by the Teacher). The 10% remaining probability was reserved for the third image of the set, which corresponds to the message “Learning is done for today, you can go home now”, which indicates that sequence of tests in class can be ended, and invites the Learner to switch from Context C back to Context A.

The volume of noise in the images in experiments evaluating the robustness of the Learner’s recognition ability was varied from 0% to 50% in steps of 5%. This percent values indicate the volume of flipped bits in the noisy images being presented to the Learner agent for recognition, with respect to the original clean image without any noise added. The total number of recognition actions performed by the RPEs associator during each performed simulation was at least 1000, in order to allow sufficient data collection for the calculation of rates of failures and successes in recognition of the shape images and Teacher guidance images.

Table II presents the rates of failures in recognizing images, for both types of images, i.e., shape forms (Contexts A and B, and images shown in Fig. 6) and Teacher’s guidance images (Context C, and images in Fig. 7). It shows the performance measures for the different volumes of image noise considered.

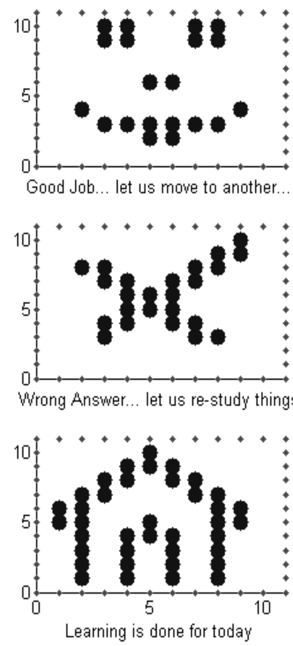

Fig. 7. Set of input images relevant for the context C, recognition of teacher guidance / feedback, produced by the Teacher after each recognition in classroom (under supervision) is concluded. The first image conveys approval, the second indicates a mistake and the third indicates “Learning is done for today, you can go home”.

The data presented in Table II indicates that only for noise levels limited to 15% we have perfect recognition of images in all the three contexts. Beyond that level of input noise, we have that the recognition of shapes (contexts A and B) happens with virtually no mistakes (bellow 1% mistakes) up to the level of input noise 20%, while the recognition of Teacher guidance images happens with virtually no mistakes up to the level of input noise 30%. We can also observe that in the range 0-30% of input noise, the failure in shape recognition can be 10 times larger (or even more) than the failure rate for Teacher’s guidance signs.

TABLE II
FAILURE RATE OF RPEs ASSOCIATORS

Input Noise Level	Failures Rates in Shape Images Recovery	Failures in Teacher’s Signs Images Recovery
0-15%	0%	0%
20%	.6%	0%
25%	1%	.1%
30%	5%	.5%
35%	14%	3%
40%	31%	13%
45%	60%	45%
50%	84%	76%

The rows show failure rates for experiments with fixed amount of input noise, shown in the first column.

This scenario of very different rates for the set of shape images and the set of Teacher guidance signals indicates the strong convenience of decoupling the two main scenarios of associations (Contexts A and B from one part, and Context C from the other), through the use of context dependent associators. This ensures that very different failure rates associated to significantly different volumes of images involved in two training sets are not mixed in a single associator. That mixture of the two sets of images would produce drastic loss of reliability in Learner's ability to follow the Teacher guidance. Through the obtained decoupling between the rates of failures in contexts of shape recognition and the rates of failures in contexts of Teacher guidance, which becomes possible with the use of context dependent associators, we are able to keep the second of these rates comparatively low, and therefore ensure that any needed re-learning induced by the Teacher, or any other guidance produced by the Teacher during learning sections, will only suffer marginally with the presence of noise in the guidance images received by the Learner.

In addition to the performance information on recognition, we also measured the distribution of recognition actions among the three relevant contexts. This result is presented in Table III, and it roughly indicates how much time the Learner spends in each of the three contexts, A, B and C. Notice that these relative frequencies of recognition actions among the three contexts depend on the specific assumptions that were done with respect to the number of shape and guidance images involved, as well as with respect to their relative probabilities. Theoretically, these relative frequencies correspond to the equilibrium configuration of a stochastic machine with three states (A, B, and C), and predefined probabilities of transition between these states.
With respect to the experimentally observed long term distribution of visits to each of the three contexts, Table III indicates that for moderate levels of input noise (up to 30%), we have p(A)=18%, p(B)=46% and p(C)=36%, which are very close to the theoretical prevision. This scenario is changed though, when the input noise goes up and we have significant volume of failures in recognizing the Teacher's guidance signs. Under these circumstances, the Learner loses the ability to follow the planned changes of contexts, and the frequencies for contexts A, B and C change significantly.

TABLE III
DISTRIBUTION OF RECOGNITION TASKS IN CONTEXTS A, B AND C

Input Noise Level	Percent of recovery cycles in Context A	Percent of recover cycles in Context B	Percent of recover cycles in Context C
0-30%	18%	46%	36%
40%	20%	45%	35%
50%	32%	26%	42%

Each row indicates the distribution among contexts A, B and C, for a fixed amount of input noise, specified at the first column.

VI. CONCLUSION

The hybrid structure here proposed was demonstrated through a scenario of image recognition being performed by an autonomous agent under different alternating contexts, including the interaction with a second agent. Processes happening at different time scales were considered in such modeling, including the robust recovery of noisy images through the blend of ordered and chaotic dynamics, as well as dialog-like sequences of visual pattern recognitions, in contexts of interaction with the Teacher agent. In this scenario, it is important to have high reliability in the pattern recognition tasks related to the Learner-Teacher dialog, and the technique of context dependent pattern recognition becomes very useful, acting as an important toll for a focused recognition. The model of alternation between unsupervised and supervised recognitions with Teacher interaction that was described in Section III and Fig. 4 can be explored, with proper adaptations, as a platform for the study of incremental learning as well as for the study of interaction and mutual training in groups of adaptive autonomous agents.

ACKNOWLEDGMENT

The authors thank the University of São Paulo, CNPq and FAPESP for supporting this research work in several forms.

REFERENCES

[1] W. Gerstner and W. M. Kistler, “Spiking Neuron Models: Single Neurons, Populations, Plasticity”, Cambridge University Press, Cambridge, UK, 2002.
[2] L. Perlovsky and R. Kozma, “Neurodynamics of higher level cognition and consciousness”, Springer-Verlag Berlin, 2007.
[3] K. Aihara, T. Takabe, and M. Toyoda, “Chaotic Neural Networks”, Physica Lett. A, vol. 144(6,7), pp. 333-340, 1990.
[4] K. Kaneko and I. Tsuda, “Complex Systems: Chaos and Beyond”, Springer-Verlag, Berlin, 2001.
[5] M. Adachi and K. Aihara, “Associative Dynamics in a Chaotic Neural Network”, Neural Networks, Elsevier, vol. 10, pp. 83–98, 1997.
[6] E. Del-Moral-Hernandez, “Non-Homogeneous Neural Networks with Chaotic Recursive Nodes: Connectivity and Multi-assemblies Structures in Recursive Processing Elements Architectures”, Neural Networks, Elsevier, vol. 18, pp. 532–540, 2005.
[7] E. Del-Moral-Hernandez, “Chaotic Searches and Stable Spatio-temporal Patterns as a Naturally Emergent Mixture in Networks of Spiking Neural Oscillators with Rich Dynamics”. In: Int. Joint Conference on Neural Networks 2006, Vancouver, Canada, 2006.
[8] W. J. Freeman, “Tutorial on neurobiology: from single neuron to brain chaos”, International Journal of Bifurcation and Chaos, vol. 2(3), pp. 451-482, 1992.
[9] E. Del-Moral-Hernandez, "Neural networks with chaotic recursive nodes: techniques for the design of associative memories, contrast with Hopfield architectures, and extensions for time-dependent inputs", Neural Networks, Elsevier, vol. 16, pp. 675–682, 2003.
[10] E. Del-Moral-Hernandez, “RPE-BAM Networks: Bidirectional Heteroassociation in Neural Networks of Recursive Nodes with Rich Dynamics”. In: International Joint Conference on Neural Networks 2008, Hong Kong, China, 2008.
[11] S. Russel, and P. Norvig, Artificial Intelligence: a modern approach, 2nd ed., Prentice Hall, 2002.
[12] J. E. Hopcroft, R. Motwani, and J. D. Ullman, *Introduction to Automata Theory, Languages, and Computation*. Addison Wesley, 2nd ed, 2006.
[13] R. C. Hilborn, "Chaos and Nonlinear Dynamics: an Introduction for Scientists and Engineers", Oxford University Press, New York, 1994.
[14] J. J. Hopfield, "Neural Networks and Physical Systems with Emergent Collective Computational Abilities", in Proc. Nat. Acad. of Sciences, USA, Ap. 1982, vol. 79, pp.2554-58.
[15] B. Kosko, "Bidirectional Associative Memories", IEEE Tran. On Syst., Man, and Cybernetics,v.18, pp.49-60, 1988.
[16] A. Carlson, and I. Fette, “Memory-based context-sensitive spelling correction at web scale,” Sixth Int. Conference on Machine Learning and Applications (ICMLA 2007), Cincinnati, Dec 2007, pp. 166-171.